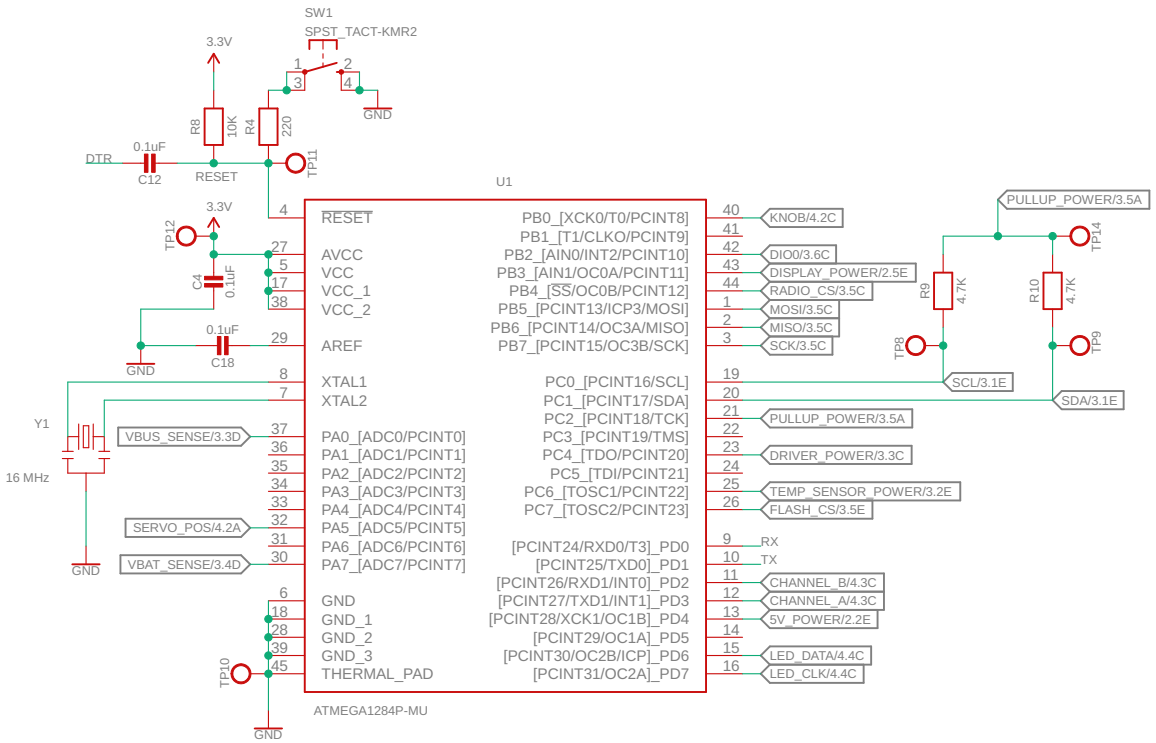
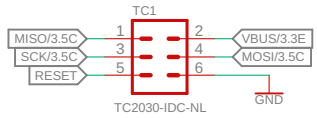
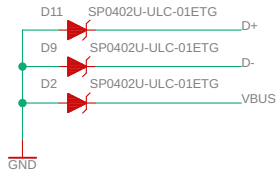
ATMEGA1284P MCU



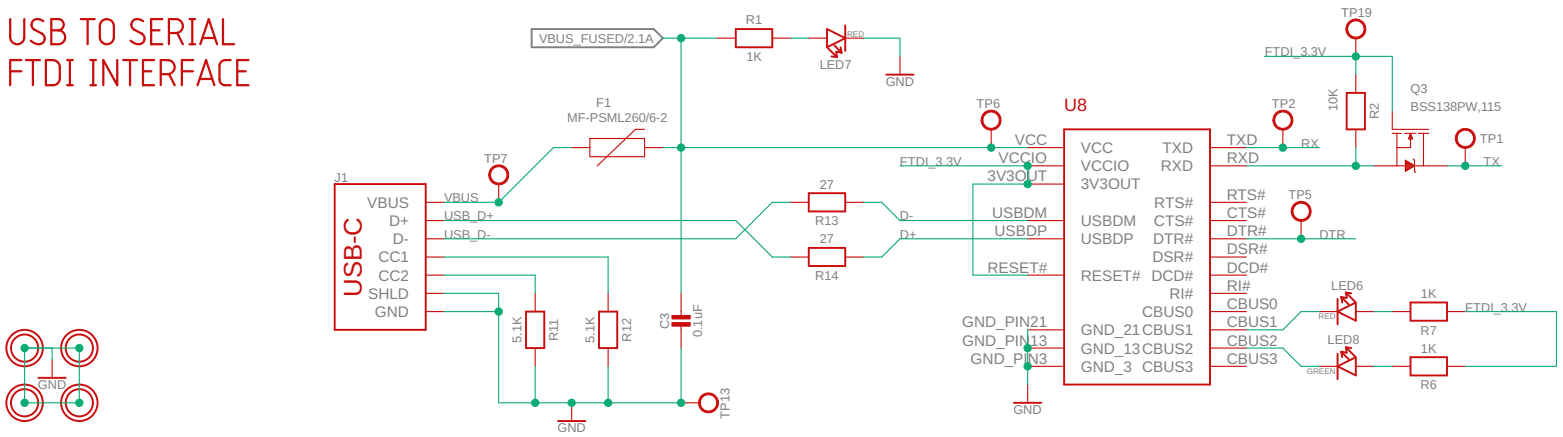
TAG-CONNECT ISP PROG HEADER



ESD DIODES



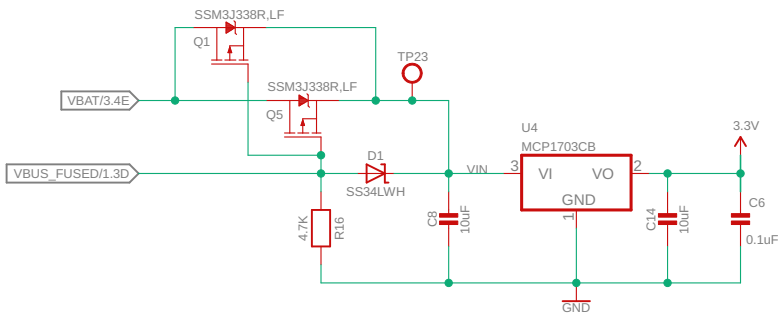
USB TO SERIAL FTDI INTERFACE



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TITLE: BORC_v2_control_board	Designer: Akram S. Ali
Sheet Title: Main Processor & interfaces	Board Version: v2.4
Date: 3/9/2025 3:39 AM	Sheet: 1/4

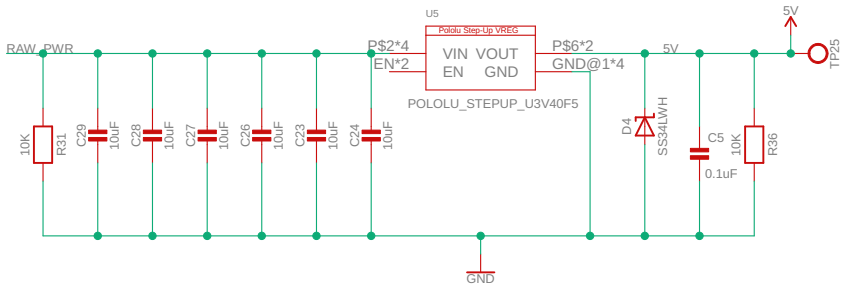
3.3V VOLTAGE REGULATOR



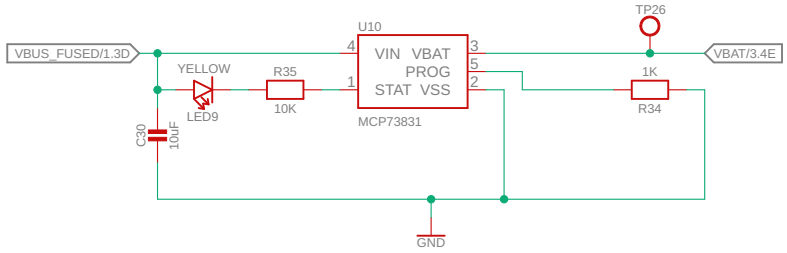
Note: Dual MOSFETs are used to lower RDS(ON) by splitting current between the two.
This effectively reduces the voltage drop across the FETs when turned on.

5V VOLTAGE REGULATOR

Note: Output voltage can be set to 5.2V by swapping R2 (on U5) to 147k ohm.
This gives headroom to account for servo's inrush current

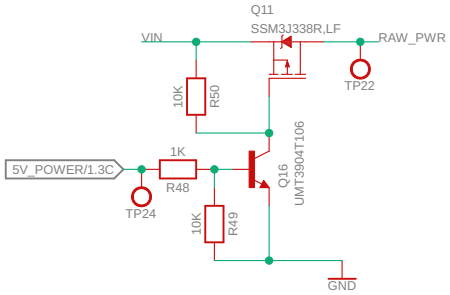


3.7V LI-ION BATTERY CHARGER

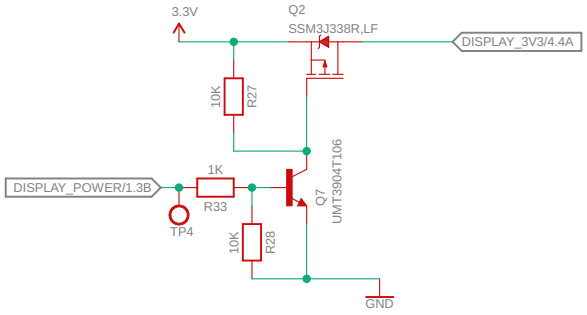


PROG Resistor:
10K ohm = 100 mA
5K ohm = 200 mA
2K ohm = 500 mA
1K ohm = 1000 mA

SERVO POWER CONTROL



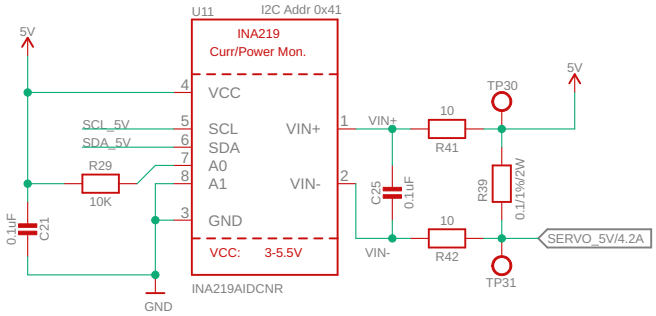
DISPLAY POWER CONTROL



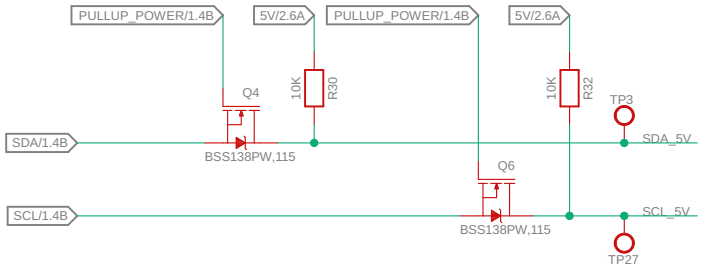
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TITLE: BORC_v2_control_board	Designer: Akram S. Ali
Sheet Title: Power Regulation & Control	Board Version: v2.4
Date: 3/9/2025 3:39 AM	Sheet: 2/4

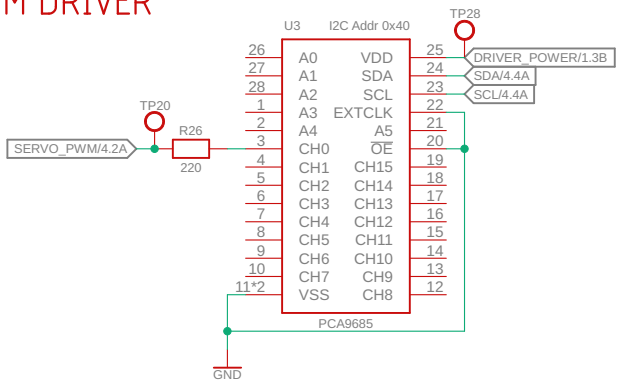
SERVO CURRENT MONITOR



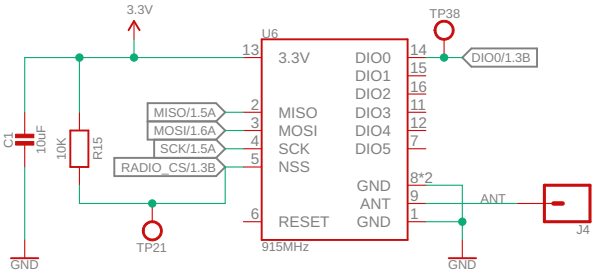
VOLTAGE LEVEL SHIFTER FOR CURRENT MONITOR



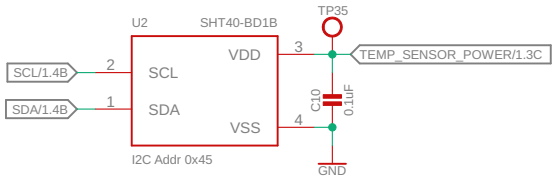
SERVO PWM DRIVER



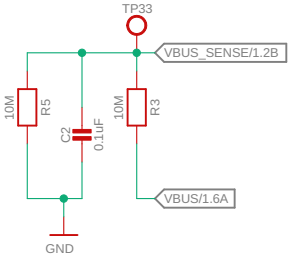
RFM69HCW RADIO



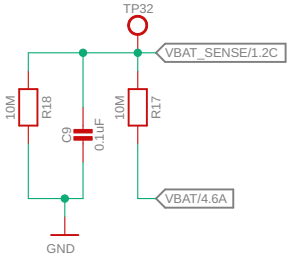
TEMPERATURE & HUMIDITY SENSOR



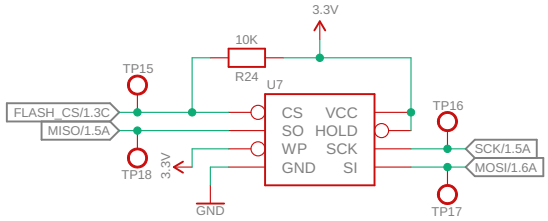
VBUS VOLTAGE MONITOR



BATTERY VOLTAGE MONITOR



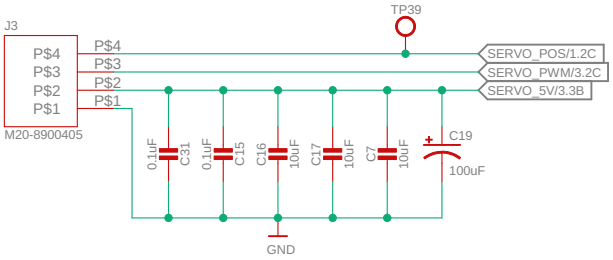
SPI FLASH MEM



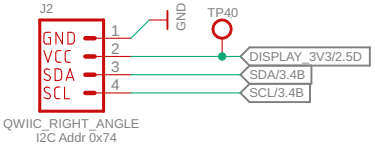
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Sheet Title: Sensors & ICs	Board Version: v2.4
Date: 3/9/2025 3:39 AM	Sheet: 3/4

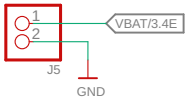
SERVO CONNECTOR



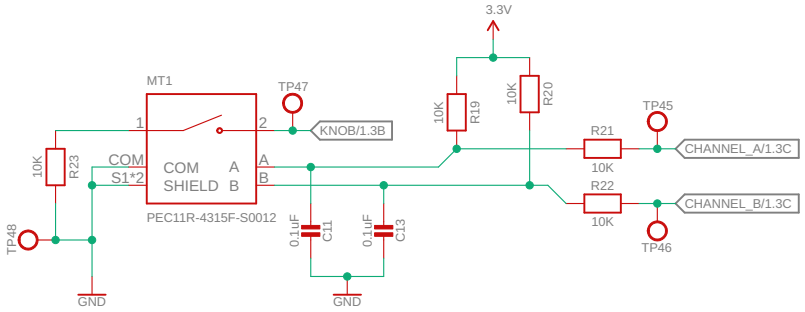
LED SCREEN CONNECTOR



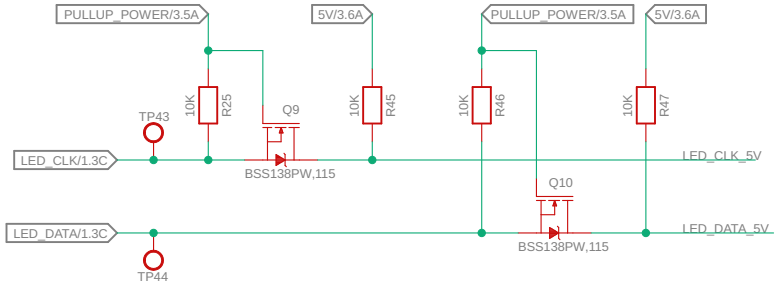
BATTERY CONNECTOR



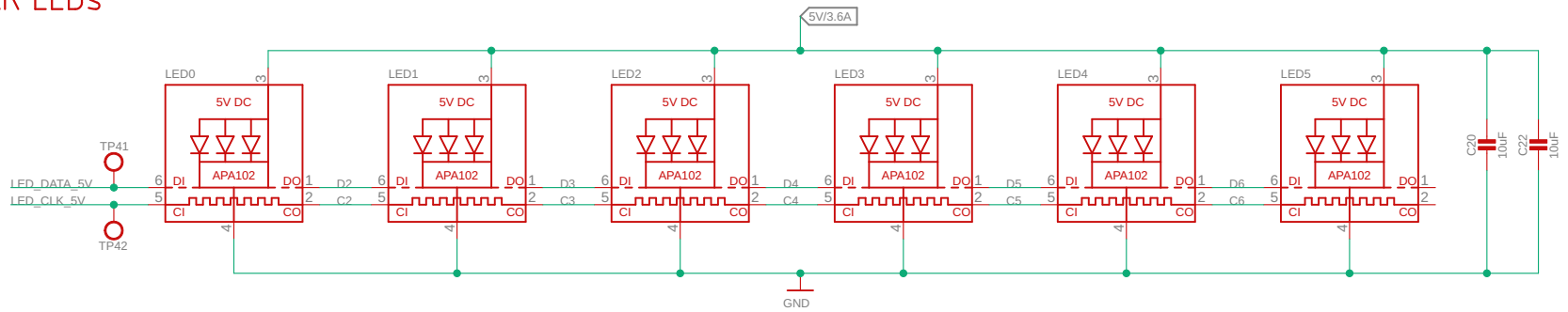
ROTARY ENCODER



VOLTAGE LEVEL SHIFTER FOR DOTSTAR LEDs



DOTSTAR LEDs



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Sheet Title: Ports, Buttons & LEDs	Board Version: v2.4
Date: 3/9/2025 3:39 AM	Sheet: 4/4